Project description: Deployment of a SIEM/XDR Solution with Wazuh

The objective of this project was to establish a centralized security monitoring environment to detect threats and manage compliance on a Windows endpoint. To achieve this, I deployed a **Wazuh Manager** on a Linux server and configured a remote agent to collect and analyze security events in real-time.

Step 1: Infrastructure Setup (Server-side)

First, I deployed an **Ubuntu Server** instance using Oracle VirtualBox, configuring the network adapter in **Bridged Mode** to ensure local connectivity. I then executed the Wazuh installation script to stand up the Indexer, Server, and Dashboard.

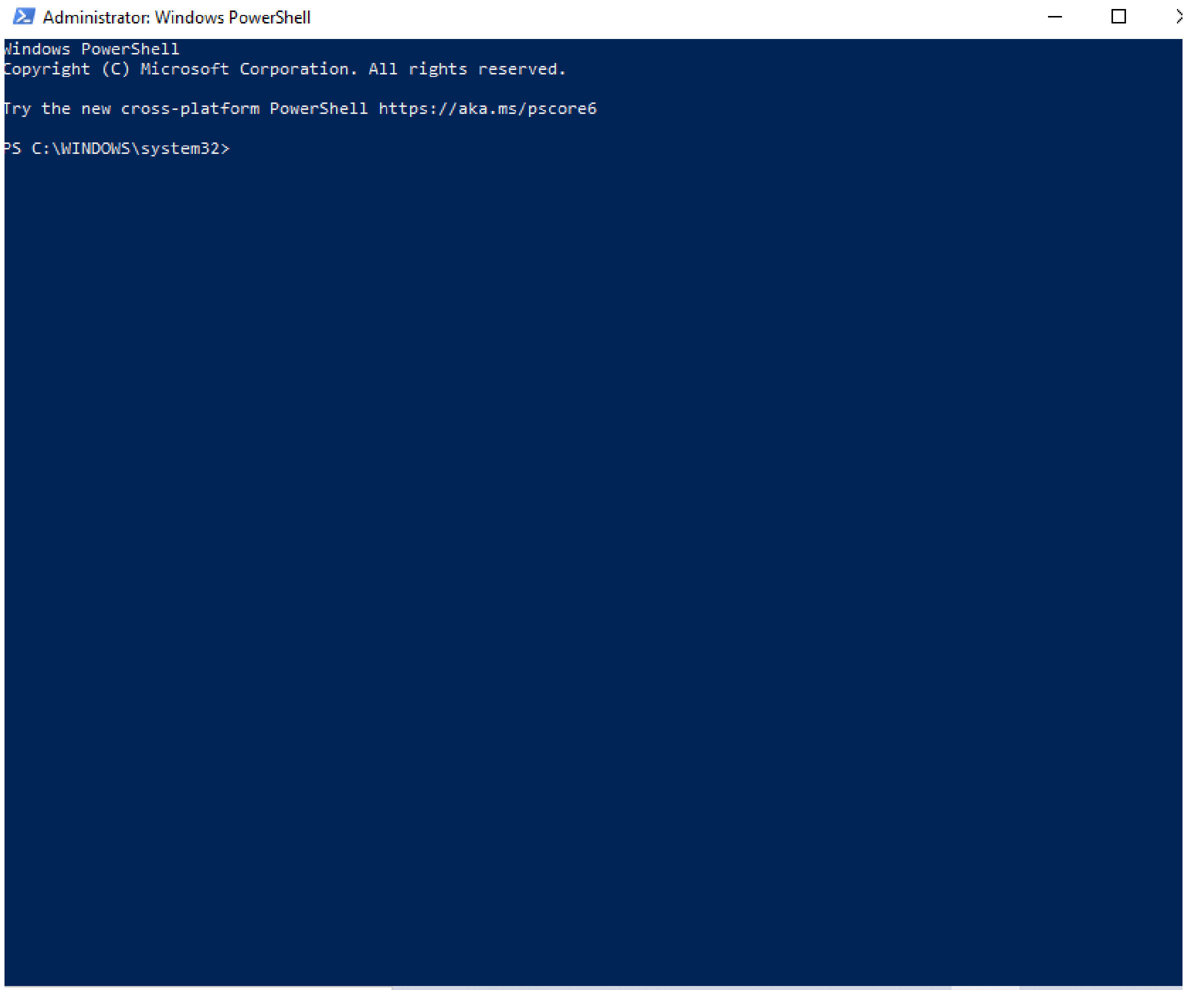
IP 192.168.X.XXX

This setup provides the central "brain" of the SIEM, where all logs will be processed and visualized through the web interface.

Step 2: Agent Deployment and Troubleshooting

To monitor my host machine, I installed the **Wazuh Agent for Windows**. Initially, the agent failed to report to the manager due to a configuration mismatch.

I used **Windows PowerShell (Admin)** to manually register the agent: & "C:\Program Files (x86)\ossec-agent\agent-auth.exe" -m **192.168.X.XXX**


Administrator: Windows PowerShell
Windows PowerShell
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Try the new cross-platform PowerShell https://aka.ms/pscore6
PS C:\WINDOWS\system32>

Administrator: Windows PowerShell

```
Windows PowerShell
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Try the new cross-platform PowerShell https://aka.ms/pscore6

PS C:\WINDOWS\system32> & "C:\Program Files (x86)\ossec-agent\agent-auth.exe" -m [redacted]
2026/01/23 16:07:35 agent-auth: INFO: Started (pid: 5972).
2026/01/23 16:07:35 agent-auth: INFO: Requesting a key from server: [redacted]
2026/01/23 16:07:58 agent-auth: ERROR: (1208): Unable to connect to enrollment service at '[[redacted]]:1515'
PS C:\WINDOWS\system32> & "C:\Program Files (x86)\ossec-agent\agent-auth.exe" -m [redacted]
2026/01/23 16:12:27 agent-auth: INFO: Started (pid: 12168).
2026/01/23 16:12:27 agent-auth: INFO: Requesting a key from server: [redacted]
2026/01/23 16:12:27 agent-auth: INFO: No authentication password provided
2026/01/23 16:12:27 agent-auth: INFO: Using agent name as [redacted]
2026/01/23 16:12:27 agent-auth: INFO: Waiting for server reply
2026/01/23 16:12:28 agent-auth: INFO: Valid key received
PS C:\WINDOWS\system32> net start wazuh-agent
The service name is invalid.

More help is available by typing NET HELPMSG 2185.

PS C:\WINDOWS\system32> Start-Service -Name "Wazuh"
PS C:\WINDOWS\system32> Start-Service -Name "wazuh-agent"
Start-Service : Cannot find any service with service name 'wazuh-agent'.
At line:1 char:1
+ Start-Service -Name "wazuh-agent"
+ ~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~
    + CategoryInfo          : ObjectNotFound: (wazuh-agent:String) [Start-Service], ServiceCommandException
    + FullyQualifiedErrorId : NoServiceFoundForGivenName,Microsoft.PowerShell.Commands.StartServiceCommand

PS C:\WINDOWS\system32> net start Wazuh
The Wazuh service is starting.
The Wazuh service could not be started.

The service did not report an error.

More help is available by typing NET HELPMSG 3534.

PS C:\WINDOWS\system32>
```
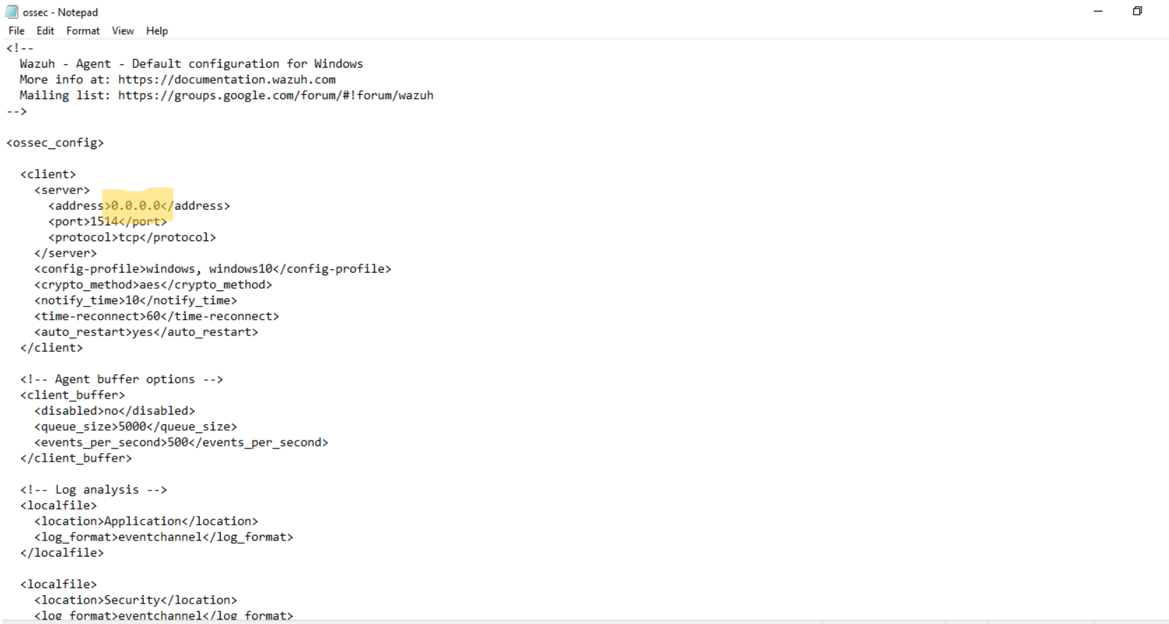

Handling Technical Issues

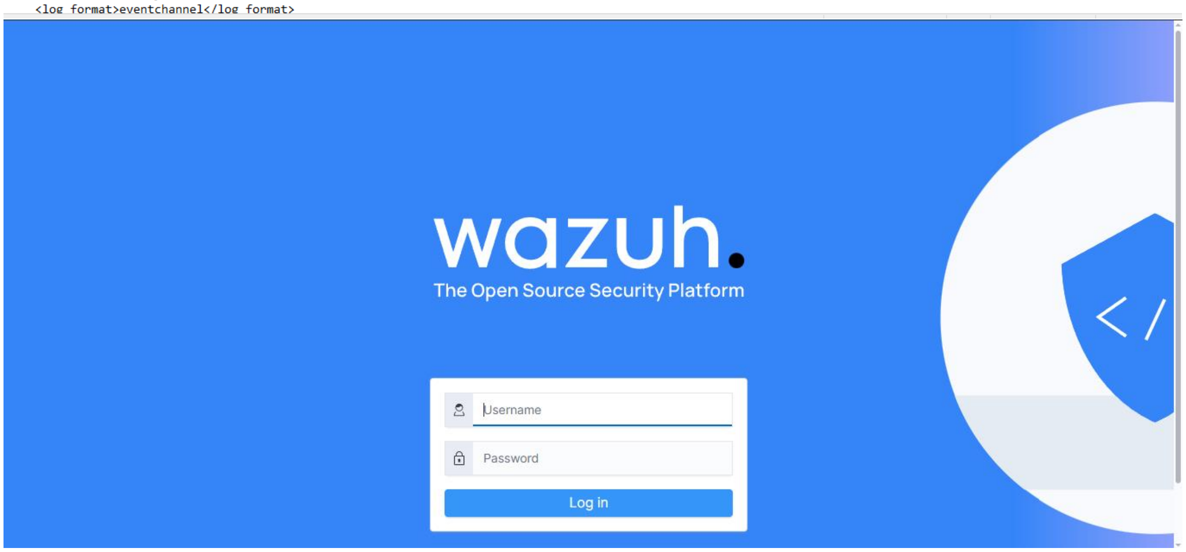
I encountered an issue where the agent service wouldn't start. Upon inspecting the ossec.conf file, I discovered the manager address was set incorrectly to 0.0.0.0. I manually edited the file using **Notepad (Admin)** to point to the correct IP: **192.168.X.XXX**



ossec - Notepad

File Edit Format View Help

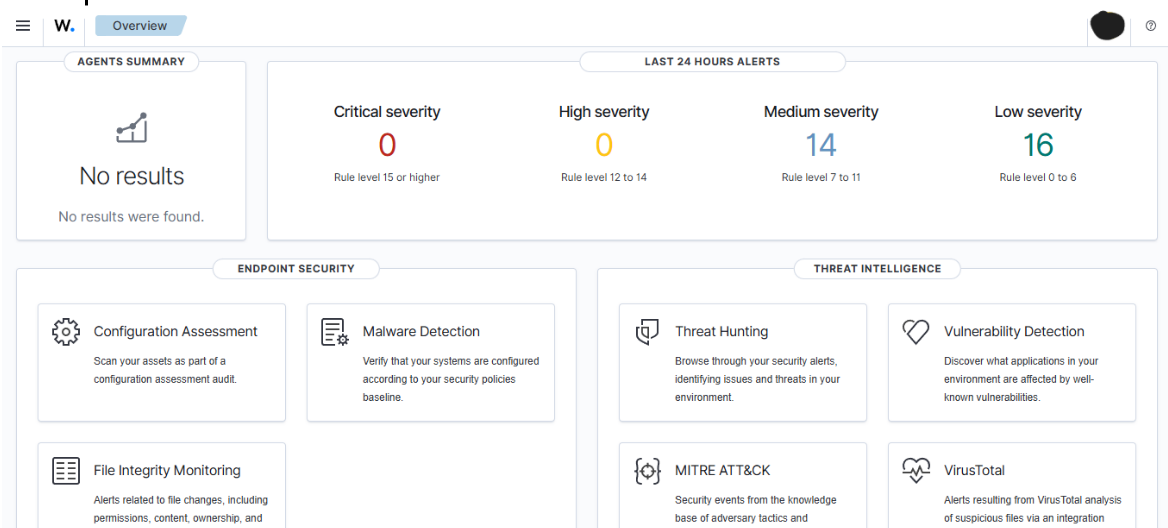
```
<!--
  Wazuh - Agent - Default configuration for Windows
  More info at: https://documentation.wazuh.com
  Mailing list: https://groups.google.com/forum/#!forum/wazuh
-->

<ossec_config>

  <client>
    <server>
      <address>0.0.0.0</address>
      <port>1514</port>
      <protocol>tcp</protocol>
    </server>
    <config-profile>windows, windows10</config-profile>
    <crypto_method>aes</crypto_method>
    <notify_time>10</notify_time>
    <time-reconnect>60</time-reconnect>
    <auto_restart>yes</auto_restart>
  </client>

  <!-- Agent buffer options -->
  <client_buffer>
    <disabled>no</disabled>
    <queue_size>5000</queue_size>
    <events_per_second>500</events_per_second>
  </client_buffer>

  <!-- Log analysis -->
  <localfile>
    <location>Application</location>
    <log_format>eventchannel</log_format>
  </localfile>

  <localfile>
    <location>Security</location>
    <log_format>eventchannel</log_format>
```

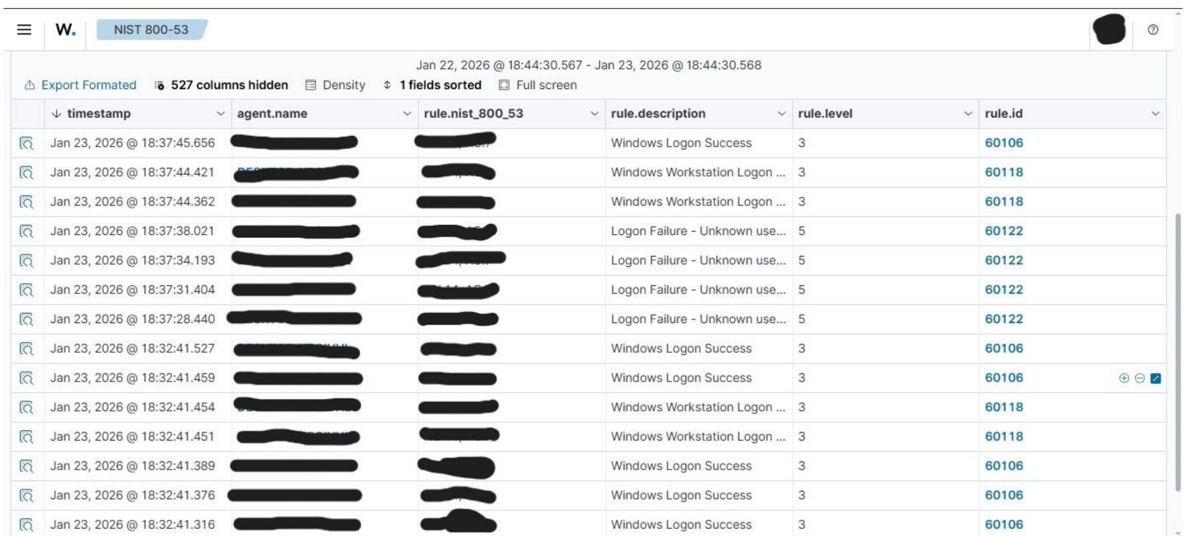


Step 3: Real-time Threat Detection (Brute Force Simulation)

To validate the deployment, I simulated a **Brute Force attack** by intentionally entering incorrect credentials multiple times on the Windows login screen.

The SIEM successfully captured these events, escalating the alert severity as the attempts continued.





timestamp	agent.name	rule.nist_800_53	rule.description	rule.level	rule.id
Jan 23, 2026 @ 18:37:45.656	[redacted]	[redacted]	Windows Logon Success	3	60106
Jan 23, 2026 @ 18:37:44.421	[redacted]	[redacted]	Windows Workstation Logon ...	3	60118
Jan 23, 2026 @ 18:37:44.362	[redacted]	[redacted]	Windows Workstation Logon ...	3	60118
Jan 23, 2026 @ 18:37:38.021	[redacted]	[redacted]	Logon Failure - Unknown use...	5	60122
Jan 23, 2026 @ 18:37:34.193	[redacted]	[redacted]	Logon Failure - Unknown use...	5	60122
Jan 23, 2026 @ 18:37:31.404	[redacted]	[redacted]	Logon Failure - Unknown use...	5	60122
Jan 23, 2026 @ 18:37:28.440	[redacted]	[redacted]	Logon Failure - Unknown use...	5	60122
Jan 23, 2026 @ 18:32:41.527	[redacted]	[redacted]	Windows Logon Success	3	60106
Jan 23, 2026 @ 18:32:41.459	[redacted]	[redacted]	Windows Logon Success	3	60106
Jan 23, 2026 @ 18:32:41.454	[redacted]	[redacted]	Windows Workstation Logon ...	3	60118
Jan 23, 2026 @ 18:32:41.451	[redacted]	[redacted]	Windows Workstation Logon ...	3	60118
Jan 23, 2026 @ 18:32:41.389	[redacted]	[redacted]	Windows Logon Success	3	60106
Jan 23, 2026 @ 18:32:41.376	[redacted]	[redacted]	Windows Logon Success	3	60106
Jan 23, 2026 @ 18:32:41.316	[redacted]	[redacted]	Windows Logon Success	3	60106

The system automatically categorized these events under international frameworks such as **NIST 800-53** (Controls AU.14 and AC.7) and **TSC**, providing immediate context for security auditors.